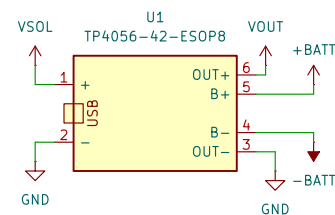


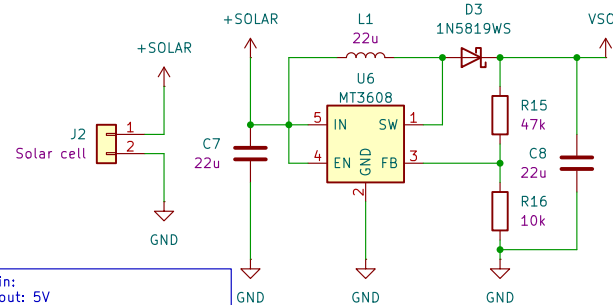
Charger



Charger Voltage: 5V
Vin: 4-8V
Charging current: 1A
Charging Cutoff: 4.2V
Discharging Cutoff: 2.4V
Output: 2.75-4.2V

Power Protection
With USB and Solar,
Shottkey should
protect Solar from the
USB, and the solar
shouldn't add up the
voltage

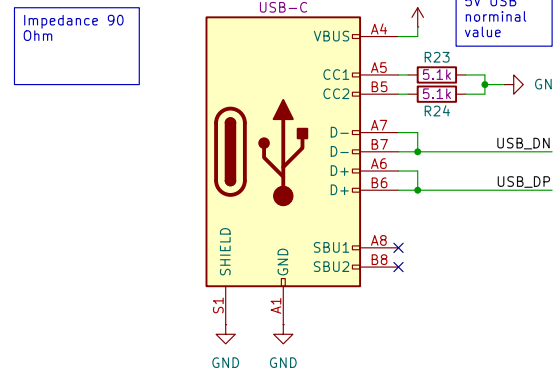
Solar Boost



Vin:
Vout: 5V
EN High voltage: 1.5V
Under Voltage Lockout Max: 2V

Output Voltage given by:
 $V_{out} = 0.6V \cdot (1 + R1/R2)$
 $5V = 0.6V \cdot (1 + R1/10k)$
 $R1 = (5V - 0.6V) \cdot 10k = 44k$

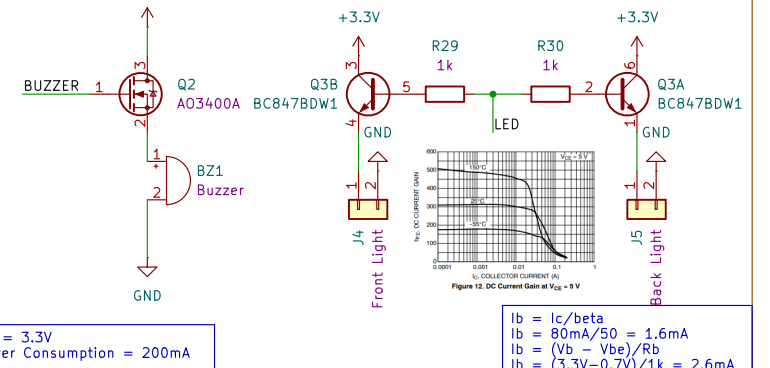
USB Programming



Impedance 90 Ohm

5V USB
nominal
value

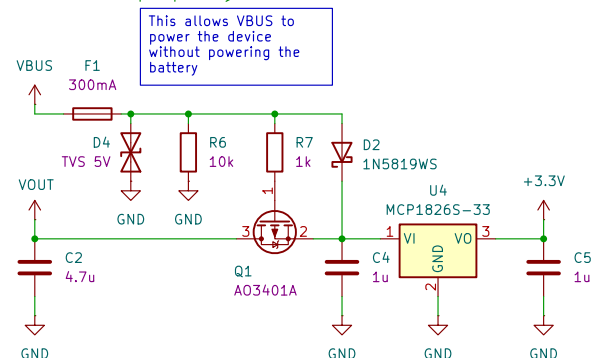
Light and Buzzer



Vin = 3.3V
Power Consumption = 200mA

$I_b = I_c/\beta$
 $I_b = 80mA/50 = 1.6mA$
 $I_b = (V_b - V_{be})/R_b$
 $I_b = (3.3V - 0.7V)/1k = 2.6mA$

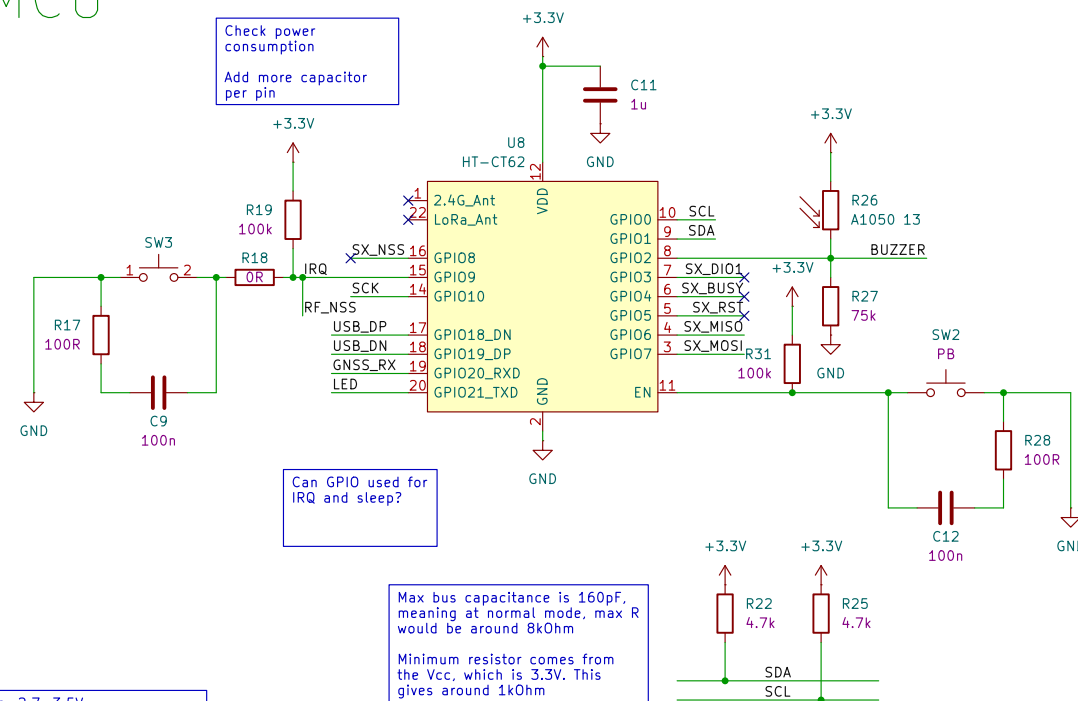
Power Supply



Max Vin: 6.5V
Vin: 2.75-4.2V
Dropout Voltage: 250mV typ, 400mV max
Vout: 3.3V
Saturation current:
Output Current: 1000mA

This allows VBUS to
power the device
without powering the
battery

MCU



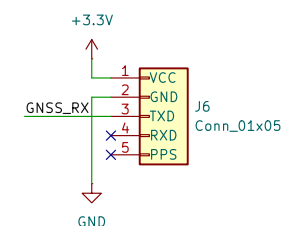
Check power
consumption
Add more capacitor
per pin

Can GPIO used for
IRQ and sleep?

Max bus capacitance is 160pF,
meaning at normal mode, max R
would be around 8kOhm
Minimum resistor comes from
the Vcc, which is 3.3V. This
gives around 1kOhm

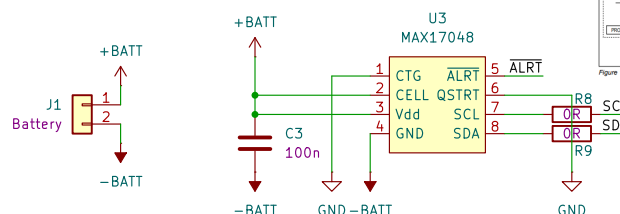
Vin: 2.7-3.5V
Power Consumption: 200mA

GNSS Connector

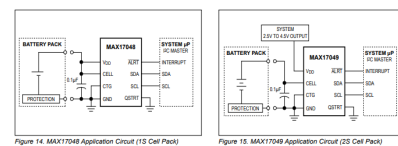


Vin = 3.3V
Baudrate = 9600
Power Consumption = 100mA

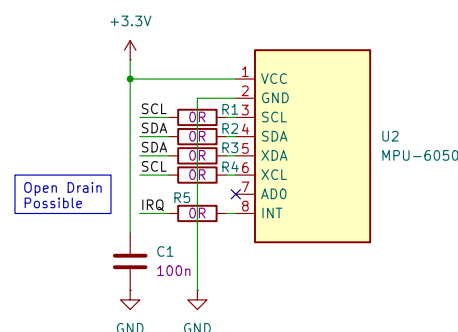
Fuel Gauge



Vin: Battery
Power Consumption: 20mA
Bus Capacitance: 120pF

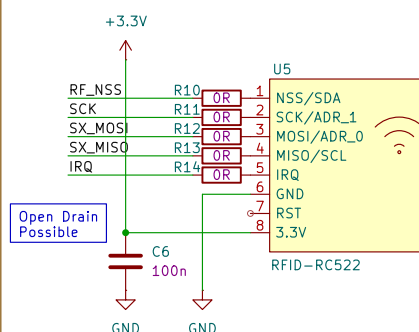


Accelerometer



Vin: 2.375-3.46V
Power Consumption: 5mA
Bus Capacitance: 20pF

RFID



Vin: 2.5-3.6V
Power Consumption: 170mA MAX

DTU Group 1

Sheet: /
File: BikeLight.kicad_sch

Title: **BeamBuddy**

Size: A3 Date: 2025-05-24

Rev:
Id: 1/1